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$$\begin{aligned}\int \frac{dx}{x(x-2)} &= \frac{A}{x} + \frac{B}{(x-2)} \\ \frac{1}{x} \Big|_{x=2} &= \frac{1}{2} = B \\ \frac{1}{(x-2)} \Big|_{x=0} &= \frac{-1}{2} = A \\ \Rightarrow \int \frac{-\frac{1}{2}dx}{x} + \int \frac{\frac{1}{2}dx}{x} \\ &= -\frac{1}{2} \ln|x| + \frac{1}{2} \ln|x-2| + C\end{aligned}$$

References